



Chapter 6

Manipulating Files How to open a Text File

File handling in Vb.net

- A file is a collection of data stored in a disk with a specific name and a directory path.
- When a file is opened for reading or writing, it becomes a stream.
- The stream is basically the sequence of bytes passing through the communication path.
- There are two main streams: the input stream and the output stream.
- The input stream is used for reading data from file read operation and
- the output stream is used for writing into the file ²

Important Libraries for the file handling

- The System.IO namespace has various classes that are used for performing various operations with files, like creating and deleting files, reading from or writing to a file, closing a file, etc.

The following table shows some commonly used non-abstract classes in the System.IO namespace:

I/O Class	Description
BinaryReader	Reads primitive data from a binary stream.
BinaryWriter	Writes primitive data in binary format.
BufferedStream	A temporary storage for a stream of bytes.
Directory	Helps in manipulating a directory structure.
DirectoryInfo	Used for performing operations on directories.
DriveInfo	Provides information for the drives.
File	Helps in manipulating files.
FileInfo	Used for performing operations on files.
FileStream	Used to read from and write to any location in a file.
MemoryStream	Used for random access of streamed data stored in memory.
Path	Performs operations on path information.
StreamReader	Used for reading characters from a byte stream.
StreamWriter	Is used for writing characters to a stream.
StringReader	Is used for reading from a string buffer.
StringWriter	Is used for writing into a string buffer.

The FileStream Class

- The FileStream class in the System.IO namespace helps in reading from, writing to and closing files.
- This class derives from the abstract class Stream.
- You need to create a FileStream object to create a new file or open an existing file.
- The syntax for creating a FileStream object is as follows:

```
Dim <object_name> As FileStream = New FileStream(<file_name>, <FileMode Enumerator>, <FileAccess Enumerator>, <FileShare Enumerator>)
```

For example, for creating a FileStream object **F** for reading a file named **sample.txt**:

```
Dim f1 As FileStream = New FileStream("test.dat", FileMode.OpenOrCreate, FileAccess.ReadWrite)
```

Parameter	Description
FileMode	The FileMode enumerator defines various methods for opening files. The members of the FileMode enumerator are: • Append: It opens an existing file and puts cursor at the end of file, or creates the file, if the file does not exist. • Create: It creates a new file. • CreateNew: It specifies to the operating system that it should create a new file. • Open: It opens an existing file. • OpenOrCreate: It specifies to the operating system that it should open a file if it exists, otherwise it should create a new file. • Truncate: It opens an existing file and truncates its size to zero bytes.
FileAccess	FileAccess enumerators have members: Read, ReadWrite and Write.
FileShare	FileShare enumerators have the following members: • Inheritable: It allows a file handle to pass inheritance to the child processes • None: It declines sharing of the current file • Read: It allows opening the file for reading • ReadWrite: It allows opening the file for reading and writing

```
Imports System.IO
Module Module1
Sub Main()
    Dim f1 As FileStream = New FileStream("test.dat", _
                                           FileMode.OpenOrCreate, FileAccess.ReadWrite)
    Dim i As Integer
    For i = 0 To 20
        f1.WriteByte(CByte(i))
    Next i
    f1.Position = 0
    For i = 0 To 20
        Console.Write("{0} ", f1.ReadByte())
    Next i
    f1.Close()
    Console.ReadKey()
End Sub
```

- End Module

Import Libraries for file handling:

- ▶ For file handling following system libraries are included.

```
Imports System.IO
```

```
Imports System.IO.StreamReader
```

```
Imports System.IO.StreamWriter
```

- ▶ For stream reader and writer.

Interface:

- Interface for file handling is:

Form1

FILE HANDLING

Display Contents of File

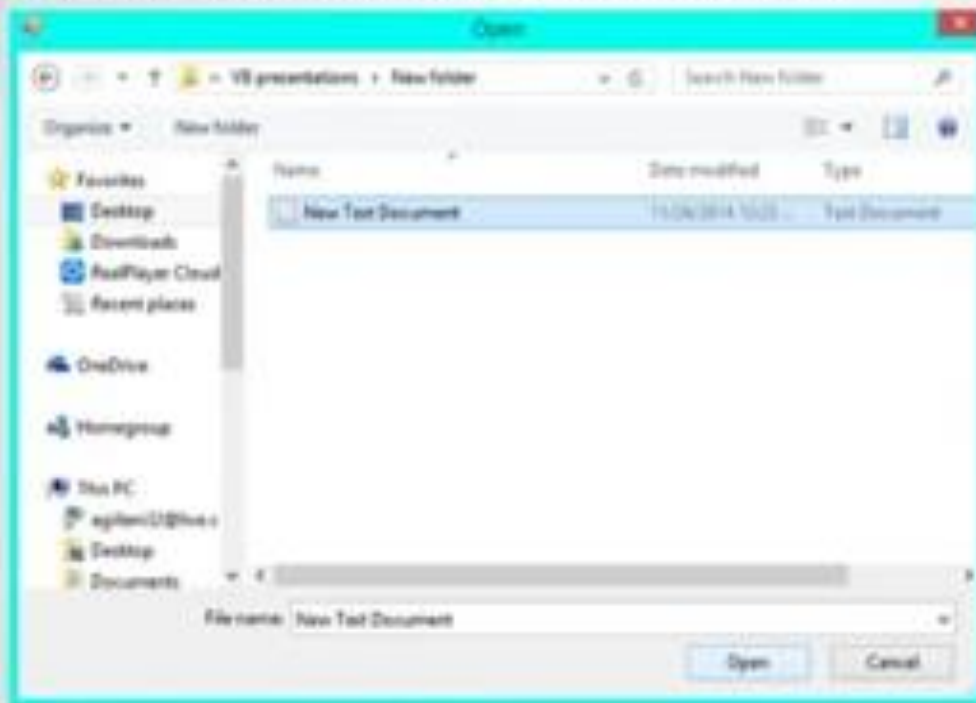
Read File
Write File
Delete File
Move File
Copy File
Append File

Clear

Open File:

Enter move/copy Path:

- By clicking on browse button following dialogue box will appear.



► The following will appear

Open File:

C:\Users\lamber\Desktop\Softwares\presentati-

 **BROWSE**

Reading a file:

For reading from a file a stream reader is used

```
Dim objReader As New StreamReader(FILE_NAME)
```

And after reading it the reading object is closed

```
objReader.Close()
```

This command reads all the contents of file to end end and displays on the richtextbox.

```
RichTextbox1.Text = objReader.ReadToEnd
```

Reading from a file (Code):

```
Private Sub Read_file_Button_Click(sender As Object, e As EventArgs) Handles Read_file_Button.Click
    Dim FILE_NAME As String = TextBox1.Text
    If (TextBox1.Text <> "") Then
        If System.IO.File.Exists(FILE_NAME) = True Then
            Dim objReader As New StreamReader(FILE_NAME)
            RichTextBox1.Text = objReader.ReadToEnd
            MsgBox("Reading Complete")
            objReader.Close()
        Else
            MsgBox("No such file exists or Invalid Path")
        End If
    Else
        MsgBox("Must select file for reading")
    End If
End Sub
```

Writing a file:

For writing streamwriter object is used.

```
Dim objWriter As New System.IO.StreamWriter(FILE_NAME)
```

For writing data from rich text box

```
objWriter.Write(RichTextBox1.Text)
```

And after that the reading object is closed

```
objWriter.Close()
```

Code:

```
Private Sub Write_file_Button_Click(sender As Object, e As EventArgs) Handles Write_file_Button.Click
    Dim FILE_NAME As String = TextBox1.Text
    If (TextBox1.Text <> "") Then
        If System.IO.File.Exists(FILE_NAME) = True Then
            Dim objWriter As New System.IO.StreamWriter(FILE_NAME)
            objWriter.Write(TextBox1.Text)
            objWriter.Close()
            MsgBox("Success! Text written to file")
            objWriter.Close()
        Else
            MsgBox("No such file exists or invalid path!")
        End If
    Else
        MsgBox("File Path must be specified for writing")
    End If
End Sub
```


Deleting a file:

- ▶ For deleting a file following command is used

```
System.IO.File.Delete(FileToDelete)
```

- ▶ Stream reader and writer is not used here.

Deletion Code:

```
Private Sub Delete_File_Button_Click(sender As Object, e As EventArgs) Handles Delete_File_Button.Click
    Dim FileToDelete As String
    FileToDelete = TextBox1.Text
    If (TextBox1.Text <> "") Then
        If System.IO.File.Exists(FileToDelete) = True Then
            System.IO.File.Delete(FileToDelete)
            MsgBox("File Deleted")
        Else
            MsgBox("no such file exists to be deleted!")
        End If
    Else
        MsgBox("Path for deleting file must be specified")
    End If
End Sub
```

Moving a file from one location to other:

- ▶ The following command is used for moving a file from one location to the other.

```
System.IO.File.Move(FileToMove, MoveLocation)
```

↘ The file to be moved

↘ The location at which the file is to be moved

Code:

```
Private Sub Move_file_Button_Click(sender As Object, e As EventArgs) Handles Move_file_Button.Click
    Dim FileToMove As String
    Dim MoveLocation As String
    FileToMove = TextBox1.Text
    MoveLocation = TextBox2.Text
    If (TextBox1.Text <> "" And TextBox2.Text <> "") Then
        If System.IO.File.Exists(FileToMove) = True Then
            System.IO.File.Move(FileToMove, MoveLocation)
            MsgBox("File Moved")
        Else
            MsgBox("No such file exists to be moved or Invalid path!")
        End If
    Else
        MsgBox("File Moving Paths must no be empty")
    End If
End Sub
```

Copying a file:

- ▶ The following command is used for copying a file from one path to the other.

```
System.IO.File.Copy
```

```
FileToCopy
```

```
NewCopy
```

*the file to be
copied*

*the path where
the file is to be
copied*

Code:

```
Private Sub Copy_file_Button_Click(sender As Object, e As EventArgs) Handles Copy_file_Button.Click
    Dim FileToCopy As String
    Dim NewCopy As String
    FileToCopy = TextBox1.Text
    NewCopy = TextBox2.Text
    If (TextBox1.Text <> "" And TextBox2.Text <> "") Then
        If System.IO.File.Exists(FileToCopy) = True Then
            System.IO.File.Copy(FileToCopy, NewCopy)
            MsgBox("File Copied")
        Else
            MsgBox("No such file exists to be copied")
        End If
    Else
        MsgBox("Copying Paths must no be empty")
    End If
End Sub
```

Moving a file from one location to other:

- ▶ The following command is used for moving a file from one location to the other.

```
System.IO.File.Move(fileToMove, newLocation)
```

↘ The file to be moved

↘ The location at which the file is to be moved

Code:

```
Private Sub Move_file_Button_Click(sender As Object, e As EventArgs) Handles Move_file_Button.Click
    Dim FileToMove As String
    Dim NewLocation As String
    FileToMove = TextBox1.Text
    NewLocation = TextBox2.Text
    If (TextBox1.Text <> "" And TextBox2.Text <> "") Then
        If System.IO.File.Exists(FileToMove) = True Then
            System.IO.File.Move(FileToMove, NewLocation)
            MsgBox("File Moved")
        Else
            MsgBox("No such file exists to be moved or Invalid path!")
        End If
    Else
        MsgBox("File Moving Paths must not be empty")
    End If
End Sub
```


Copying a file:

- ▶ The following command is used for copying a file from one path to the other.

```
System.IO.File.Copy(FileToCopy, NewCopy)
```

the file to be copied

the path where the file is to be copied

Code:

```
Private Sub Copy_file_Button_Click(sender As Object, e As EventArgs) Handles Copy_file_Button.Click
    Dim FileToCopy As String
    Dim NewCopy As String
    FileToCopy = TextBox1.Text
    NewCopy = TextBox2.Text
    If (TextBox1.Text <> "" And TextBox2.Text <> "") Then
        If System.IO.File.Exists(FileToCopy) = True Then
            System.IO.File.Copy(FileToCopy, NewCopy)
            MsgBox("File copied")
        Else
            MsgBox("No such file exists to be copied")
        End If
    Else
        MsgBox("Copying Paths must no be empty")
    End If
End Sub
```

Appending a file:

- ▶ Appending is as same as writing the difference is that

```
Dim objWriter As New System.IO.StreamWriter(FILE_NAME, True)
```

True indicates that the previous content remains the same and the file is opened for appending

- ▶ After that the streamwriter is closed.
- ▶ `objWriter.Close()`

Code:

```
Private Sub Append_File_Button_Click(sender As Object, e As EventArgs) Handles Append_File_Button.Click
    Dim FILE_NAME As String = TextBox1.Text
    If (TextBox1.Text <> "") Then
        If System.IO.File.Exists(FILE_NAME) = True Then
            Dim objWriter As New System.IO.StreamWriter(FILE_NAME, True)
            objWriter.WriteLine(RichTextBox1.Text)
            objWriter.Close()
            MsgBox("Success! Text Appended to the File")
            objWriter.Close()
        Else
            MsgBox("No such file exists to be appended")
        End If
    Else
        MsgBox("Must Enter File name/path to append it")
    End If
End Sub
```

Code for displaying the contents of file:

```
Private Sub Display_Contents_Button_Click(sender As Object, e As EventArgs) Handles Display_Contents_Button.Click
    Dim Reader As StreamReader
    Dim FILE_NAME As String = TextBox1.Text
    Try
        Reader = New StreamReader(FILE_NAME, True)
        RichTextBox1.Text = Reader.ReadToEnd
    Catch ex As Exception
        MsgBox(ex.ToString)
    End Try
End Sub
```

For opening a file:

- ▶ For opening a file a file dialogue box is used
- ▶ Open file dialogue box is dragged and dropped to the program.



- ▶ And following code is written:

```
Private Sub file1_open_Button_Click(sender As Object, e As EventArgs) Handles file1_open_Button.Click
    If OpenFileDialog1.ShowDialog = DialogResult.OK Then
        TextBox1.Text = OpenFileDialog1.FileName
    End If
End Sub
```